

Cement-1

Visible	Ref.Code	Score	Compound Name	Displ.[°2θ]	Scale Fac.	Chem. Formula
*	01-089-1304	57	Magnesium Calcium Carbonate	0.000	0.691	(Mg0.03 Ca0.97) (C O3)
*	00-055-0740	53	Calcium Silicate	0.000	0.403	Ca3 Si O5
*	01-083-2466	14	Silicon Oxide	0.000	0.079	Si O2
*	05-001-0450	14	Calcium Aluminum Oxide	0.000	0.094	Ca2 Al2 O5
*	01-073-2077	29	Calcium Oxide Silicate	0.000	0.308	Ca3 (Si O4) O

Compound	Scale Factor	Phase % (approx)
Magnesium Calcium Carbonate	0.691	43.87%
Calcium Silicate (Ca3 SiO5)	0.403	25.59%
Silicon Oxide (SiO2)	0.079	5.02%
Calcium Aluminum Oxide (Ca2 Al2 O5)	0.094	5.97%
Calcium Oxide Silicate (Ca3 SiO5)	0.308	19.55%

Cement-2

Visible	Ref.Code	Score	Compound Name	Displ.[°2θ]	Scale Fac.	Chem. Formula
*	00-055-0740	61	Calcium Silicate	0.000	0.685	Ca ₃ Si O ₅
*	01-072-1652	56	Calcium Carbonate	0.000	0.806	Ca (C O ₃)
*	03-065-0466	12	Silicon Oxide	0.000	0.061	Si O ₂
*	01-082-9280	10	Magnesium Iron Carbonate	0.000	0.099	(Fe _{0.65} Mg _{0.35}) (C O ₃)

Compound	Scale Factor	Phase % (Weight Approx)
Calcium Silicate (Ca ₃ SiO ₅)	0.685	41.49%
Calcium Carbonate (CaCO ₃)	0.806	48.82%
Silicon Oxide (SiO ₂)	0.061	3.69%
Magnesium Iron Carbonate	0.099	6.00%

Cement-3

Visible	Ref.Code	Score	Compound Name	Displ.[°2θ]	Scale Fac.	Chem. Formula
*	00-042-0551	61	Calcium Silicate	0.000	0.596	Ca ₃ Si O ₅
*	01-072-4582	49	Calcium Carbonate	0.000	0.525	Ca (C O ₃)
*	01-085-0865	16	Silicon Oxide	0.000	0.064	Si O ₂
*	01-080-2797	17	Calcium Carbonate	0.000	0.146	Ca (C O ₃)
*	01-083-2208	15	Calcium Aluminum Silicate	0.000	0.136	Ca ₃ Al ₂ Si ₃ O ₁₂
*	01-073-2091	18	Calcium Silicate	0.000	0.298	Ca ₂ (Si O ₄)

Compound	Scale Factor	Phase % (Approx)
Alite (Ca ₃ SiO ₅)	0.596	33.77%
Belite (Ca ₂ SiO ₄)	0.298	16.88%
Calcium Carbonate (CaCO ₃)	0.671*	38.02%
Calcium Aluminum Silicate	0.136	7.71%
Silicon Oxide (SiO ₂)	0.064	3.62%

Cement-4

Visible	Ref.Code	Score	Compound Name	Displ.[°2θ]	Scale Fac.	Chem. Formula
*	01-085-1378	36	Calcium Oxide Silicate	0.000	0.406	Ca ₃ (Si O ₄) O
*	00-011-0593	36	Calcium Magnesium Aluminum Silicate	0.000	0.458	54 Ca O · 16 Si O ₂ · Al ₂ O ₃ · Mg O
*	01-073-2091	10	Calcium Silicate	0.000	0.381	Ca ₂ (Si O ₄)
*	05-001-0450	14	Calcium Aluminum Oxide	0.000	0.182	Ca ₂ Al ₂ O ₅

Compound	Scale Factor	Phase % (Approx)
Alite (C ₃ S - Calcium Oxide Silicate)	0.406	28.45%
Complex Ca-Mg-Al Silicate	0.458	32.10%
Belite (C ₂ S - Calcium Silicate)	0.381	26.70%
Calcium Aluminum Oxide	0.182	12.75%

Process of Calculation

Compound	Scale Factor
Ca_3SiO_5	0.685
CaCO_3	0.806
SiO_2	0.061
$(\text{Fe}_{0.65}\text{Mg}_{0.35})\text{CO}_3$	0.099

Total scale factor:

$$\text{Total} = 0.685 + 0.806 + 0.061 + 0.099 = 1.651$$
$$\text{CaCO}_3 = (0.806 / 1.651) \times 100 = 48.8\%$$

Same for others....

- $\text{Ca}_3\text{SiO}_5 = 41.5\%$
- $\text{CaCO}_3 = 48.8\%$
- $\text{SiO}_2 = 3.7\%$
- $(\text{Fe}, \text{Mg})\text{CO}_3 = 6.0\%$

Example 1: CaCO_3

Atomic weights:

- $\text{Ca} = 40$
 - $\text{C} = 12$
 - $\text{O}_3 = 48$
- Total = 100

So:

- $\text{Ca} = 40\%$
- $\text{C} = 12\%$
- $\text{O} = 48\%$

Now multiply by phase % (48.8%):

- $\text{Ca} = 48.8 \times 0.40 = \mathbf{19.5\%}$
- $\text{C} = 48.8 \times 0.12 = \mathbf{5.86\%}$
- $\text{O} = 48.8 \times 0.48 = \mathbf{23.4\%}$